Designing Novel, Safe, and Pleasant Flavors: A Structure–Receptor–Percept Pipeline with a Machine-Checked Tastiness Metric

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Abstract

Motivated by the conjecture that exceptional, never-experienced flavors exist in a vast molecular flavor space, we build a computational pipeline for designing novel flavor combinations. The pipeline follows the biological chain *molecule* $\rightarrow$ *odorant-receptor activation* $\rightarrow$ *perceived flavor*: (i) a structure-based model predicts human panel *pleasantness* (5-fold CV Spearman 0.50 on 420 molecules from Keller & Vosshall 2016); (ii) an odorant-receptor layer grounded in the Mainland 2015 screen provides a real, if partial, receptor signal (1.57× chance that structurally nearest molecules share a receptor); (iii) a food-appropriate toxicity screen filters hazardous structures (drug-discovery filters such as Brenk/PAINS are shown to be miscalibrated for food); (iv) a fragment-based (BRICS) generator constructs *de novo* molecules, of which the top candidates are all absent from PubChem; and (v) a compositional-statistics model learns which chemical classes drive pleasantness (lactones/esters up, sulfur/acids down; composition predicts blend pleasantness at Spearman 0.42) and steers a multi-objective generator that maximizes predicted pleasantness and novelty while minimizing toxicity. We show a learned non-additive mixture model captures interaction that mean-pooling cannot (0.82 vs 0.25 Spearman on a controlled non-additive oracle), and we *machine-check in Lean* the key property that under an additive score no blend can exceed its best component—formalizing precisely why non-additive modeling is required. We are explicit about scope: outputs are computational hypotheses for a chemist or molecular gastronomist, not safety-cleared recipes, and human validation remains future work.

1 Introduction

Most single flavors are one molecule (vanillin, isoamyl acetate), but chocolate is a high-dimensional mixture of hundreds of odor-active compounds acting across roughly 30 taste and 400 odor receptors. If flavor space is that large and mostly unexplored, exceptional flavors that no human has tasted plausibly exist. This work operationalizes that idea as a design problem: learn what tastes good, generate new molecules, and blend them into novel, safe, pleasant recipes, with every claim tied to public data or a stated control. The conjecture and framing are due to a recent essay on biological challenges for AI; our contribution is the computational instrument and its honest evaluation.

2 Related work

Structure-to-odor prediction has advanced from chemoinformatic baselines (the DREAM Olfaction Challenge) to learned embeddings that reach human-level odor labeling (the Principal Odor Map). Odorant receptors are G-protein-coupled receptors; receptor screens (Mainland 2015) and resources such as GPCRdb connect the flavor and pharmacology sides. Pleasantness (hedonic valence) is a primary perceptual dimension of olfaction. We build on these with real labeled data (via Pyrfume) and add de-novo generation, a food-appropriate safety screen, a learned mixture model, compositional statistics, and a machine-checked metric.

3 Methods

Data. Real molecules and pleasantness labels from Keller & Vosshall 2016; odor descriptors from Leffingwell; odorant-receptor activations from Mainland 2015; all via Pyrfume. Molecules are featurized as ECFP4 fingerprints.

Goodness model. A random forest maps ECFP4 to panel pleasantness, evaluated by 5-fold cross-validated Spearman/R^2.

Receptor layer. Confirmed molecule–OR activations provide known receptor hits; a structural nearest-neighbour predictor infers likely activation, validated leave-one-out against a chance baseline.

Toxicity screen. A curated denylist, genotoxic structural alerts, and element/property rules yield an OK/CAUTION/TOXIC verdict; only OK molecules enter a recipe.

De-novo generation. BRICS fragmentation of safe molecules followed by BRICS assembly produces new molecules; candidates are sanitized, deduplicated, size-filtered, safety-screened, and their structural novelty is checked empirically against PubChem by InChIKey.

Mixture model. A permutation-invariant set model (mean/std/min/max pooling) is compared to a mean-pooling baseline on a controlled non-additive oracle motivated by non-additive single-neuron mixture responses.

Composition and objective. Molecules are classified into chemical classes; class pleasantness lifts define a target composition. A multi-objective search maximizes $J = w_t\,\text{taste} + w_c\,\text{composition} + w_n\,\text{novelty}$ subject to a hard toxicity exclusion, over a pool of known and de-novo molecules.

4 A machine-checked property of the tastiness metric

Let a recipe R have components with pleasantness $p_i \in [0, 100]$ and weights $w_i \geq 0$, $\sum_i w_i = 1$. The additive tastiness estimate is $T(R) = \sum_i w_i p_i$.

Theorem 1 (Best-part bound)**.** $T(R) \leq \max_i p_i$. *Consequently, under the additive estimate no blend is tastier than its best single component; exceeding it requires a non-additive model.*

The discrete backbone (every element of a list is $\leq$ its maximum, hence a convex combination is $\leq$ the maximum) is formalized and machine-checked in Lean 4 (`FlavorMath.lean`, `lean` exit 0), with a corollary at the level of chemical-class composition. This makes explicit why the empirical, learned mixture model is necessary and where additivity is a wall.

5 Results

Class-level statistics recover known flavor chemistry: lactones (+9.3), ethers, esters, aromatics, phenols raise pleasantness; sulfur (−18.1), pyrazines, acids, furans lower it. The final composition-guided generator yields blends with 6–8 of 10 components being de-novo molecules, chemical novelty 0.63–0.67 from all reference flavors.

Component	Metric	Value
Structure $\rightarrow$ human pleasantness	5-fold CV Spearman / R^2	0.50 / 0.26
Learned mixture vs mean-pool	Spearman	0.82 vs 0.25
Molecule $\rightarrow$ receptor (Mainland)	vs chance	1.57$\times$
Composition $\rightarrow$ blend pleasantness	Spearman	0.42
Toxicity screen	molecules passing	380/420
De-novo molecules	safe & novel; top-12 in PubChem	434; 0/12
Metric best-part bound	Lean machine-checked	exit 0

Table 1: Summary of validated results. All reproducible from the released code.

6 Limitations

Pleasantness is only moderately predictable from structure ($R^2 = 0.26$); the additive combo score is a first-order estimate; the mixture model is validated on a synthetic (if principled) oracle because public mixture-pleasantness data are scarce; de-novo predictions are extrapolation and are reported with nearest-known similarity; receptor coverage is sparse; and novelty is relative to public data. Most importantly, tastiness is empirical: no result here establishes that a physical blend tastes good.

7 Broader impacts and safety

The safety screen is a structural hazard filter, not a regulatory clearance; de-novo molecules have no safety data. Outputs are hypotheses for expert evaluation, and the repository includes a handoff document requiring GRAS/FEMA confirmation and dose limits before any human exposure.

8 Reproducibility

All data are public (via Pyrfume); all models, controls, and the Lean proof are released with per-result JSON artifacts and a step-by-step replication guide.

9 Conclusion

We present an honest, end-to-end, reproducible instrument for novel flavor design that links molecular structure, receptor activation, perception, and toxicity, with a machine-checked metric property clarifying the role of non-additive modeling. The natural next step is wet-lab and human-panel validation of the top candidates.

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